

# Mohammad Bahrami.

AMR Application Engineer — *intralogistics solutions, fleet sizing & customer deployment*

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## + SEC.01 Profile

3+ YEARS · AUTOMATION

Application Engineer with 3+ years in intralogistics automation, specialising in AMR/AGV solution design, fleet sizing, and customer deployment. Delivered 40+ feasibility studies and automation projects across Europe, translating customer requirements into technically validated, commercially viable automation proposals. Currently leading Easy Sizing as Product Owner — an internal fleet-sizing tool used by sales engineers to scope and price AMR solutions in minutes.

## + SEC.02 Experience

2022 – PRESENT

I

JAN 2025 – PRESENT

### AMR Application Engineer

DYMATION / MILAN, IT

- Lead product ownership of Easy Sizing, an internal AMR/AGV fleet-sizing tool used by sales engineers - gathering stakeholder requirements, managing solution backlog, and coordinating a 3-person development team to deliver ranked fleet configurations and pricing proposals in minutes.
- Conducted customer feasibility studies across 4 countries, assessing warehouse operations, material flow requirements, and technical constraints to validate autonomous-vehicle integration and define fleet sizing recommendations.
- Supported pre-sales activities by translating complex simulation outputs into clear business cases, enabling sales teams to present ROI-driven automation proposals to customers.

II

MAY 2023 – DEC 2024

### Simulation & Digital Twin Engineer

DYMATION / MILAN, IT

- Delivered 35+ Emulate3D warehouse simulation models for AMR/AGV deployments across Europe, reducing simulation development time from 3 months to 1 week via modular, reusable architecture.
- Performed pre-deployment system validation by integrating Siemens TIA Portal and Beckhoff TwinCAT PLC environments with Emulate3D simulations, reducing on-site commissioning time by 10%.
- Conducted fleet sizing and traffic optimisation studies for AGV/AMR deployments, achieving up to 20% fleet count reduction through material flow analysis and throughput modelling.
- Applied FMEA and Lean Six Sigma methodology to identify system bottlenecks, improving predicted throughput by 12–15%.
- Developed immersive pre-sales VR demonstrations in Emulate3D, enabling customers to visualise automation solutions and supporting the sales cycle from concept to commitment.

III

OCT 2022 – MAY 2023

### Master's Thesis Intern

DYMATION / MILAN, IT

- Developed a custom OPC UA server in C# for bi-directional data exchange between Emulate3D and control systems.
- Designed a TCP-based peer-to-peer messaging protocol for AMR-to-AMR communication, real-time traffic coordination, and deadlock prevention.
- Optimised job-management logic, improving simulated throughput by 7%; recommendations implemented in a production facility.

DEGREE.01 – MSc

Mechanical Engineering — Mechatronics & Robotics

Politecnico di Milano / MILAN, IT

FEB 2021 – MAY 2023

*Polimi Ambassador in Smart Infrastructures.* Thesis — Virtual Commissioning for Automatic Systems in Intralogistics. developed a custom OPC UA server and AMR-to-AMR communication protocol, and recommendations were adopted in a production facility.

DEGREE.02 – BSc

Mechanical Engineering

Amirkabir University of Technology / TEHRAN, IR

SEP 2016 – AUG 2020

*Teaching Assistant — Heat Transfer.* Rotating quiz and problem-solving sessions. Publication — doi:10.1016/J.ICHEATMASSTRANSFER.2021.105360

01 Simulation & Digital Twin

Emulate 3D

Virtual Commissioning

Discrete Event Simulation

Digital-Twin Modelling

02 Fleet & Intralogistics

AGV / AMR Fleet sizing

Traffic Optimisation

Material-Flow Analysis

Warehouse Automation

03 Programming & Tools

Git / GitHub

C#

Linux / CLI

Python

REST API

SQL & Database

04 Practice

Product Ownership

Project Management

Lean Six Sigma

Risk Analysis

Stakeholder Management

FMEA

Lean Six Sigma — Yellow Belt

6SigmaStudy

PROJECT MANAGEMENT - Prince2

PRINCE2

Machine Learning

COURSERA

Persian

NATIVE

English

C1

Italian

B1

German

A1 · LEARNING